

Introduction to AI & Machine Learning in AIoT



Understand AI fundamentals in practical language

Understand how ML works conceptually

Understand how AI integrates into IoT systems

Build their first AIoT ML workflow

Perform real-time predictions

Understand the architecture mindset behind AIoT systems

What is AI

Artificial Intelligence (AI)

- Broad field of computer science dedicated to building software systems capable of performing tasks that typically require human intelligence
- These tasks include reasoning, learning from past experiences, making decisions, recognizing speech, and understanding visual information

Everyday Examples of AI



Generative AI: Tools like ChatGPT or Gemini that write essays, code, and brainstorm ideas.



Recommendation Engines: Netflix or Spotify algorithms predicting what you want to watch or hear next.



Computer Vision: Your smartphone unlocking via FaceID, or a Tesla detecting a stop sign.



Natural Language Processing: Smart assistants like Alexa or Siri understanding your spoken commands.

AI (The Big Goal):

- The overall concept of making machines smart.

Machine Learning (The Method):

- A specific subset of AI focused on letting computers learn from data without explicit programming.

Model (The Engine):

- The actual, functional asset created by Machine Learning that powers the AI application.

Perception:

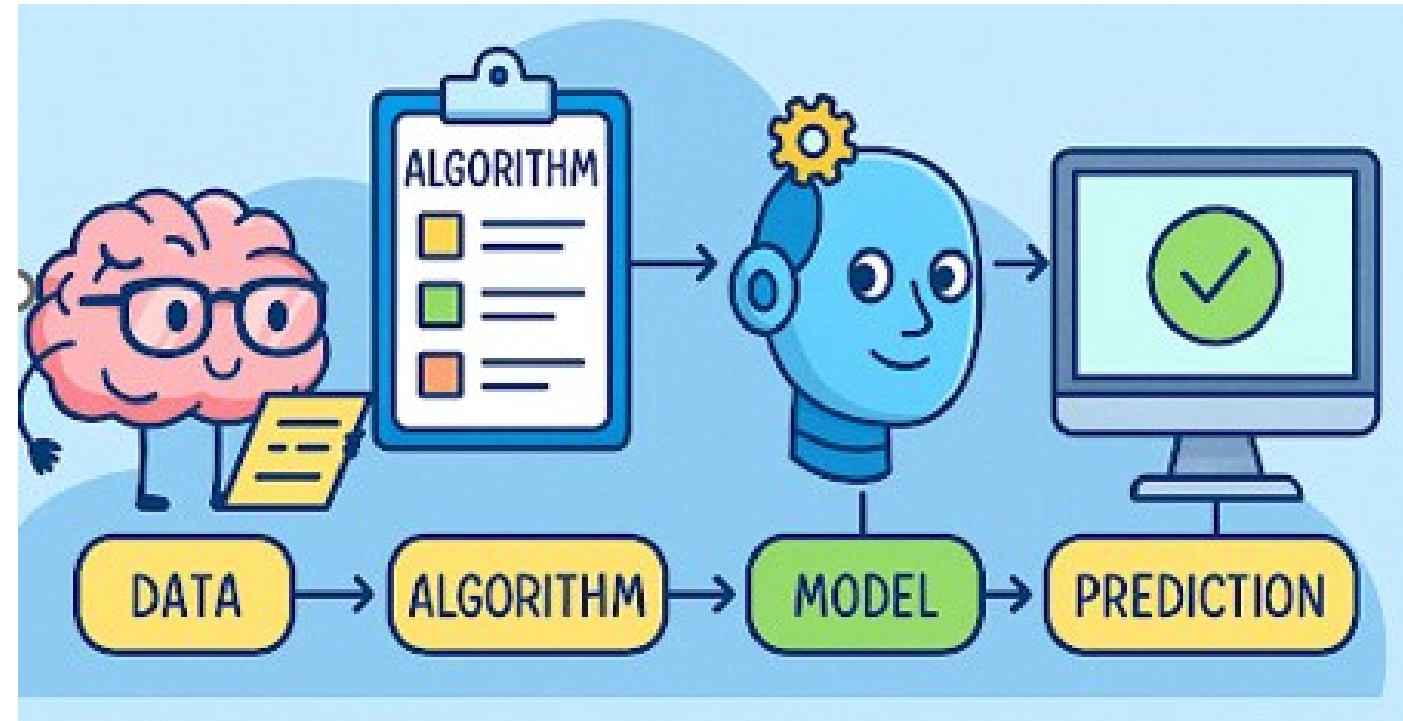
Understanding the world through sight (computer vision) or sound (audio processing).

Cognition:

Analyzing data, recognizing patterns, solving complex problems, and making predictions.

Action: Generating new text, creating images, coding, or physically moving a robotic arm.

How AI works



- Model is core engine of AI which is used for prediction
- Model is trained on huge datasets using Machine Learning techniques
- Model learns by recognizing patterns and relations of variables in data
- Once Trained the model is used to make prediction on new data

Model = Algorithm + Data + Training

Algorithm – Logic or mathematical recipe. It defines how the system will learn, but it doesn't know anything yet. Examples: Neural Networks, Decision Trees, or Linear Regression.

Data - raw information that the algorithm studies. It includes examples, facts, images, or text. Without data, the algorithm has nothing to learn from.

Training / Learning - This is the active process where the algorithm runs through the data over and over. It makes mistakes, corrects itself, and adjusts its internal math

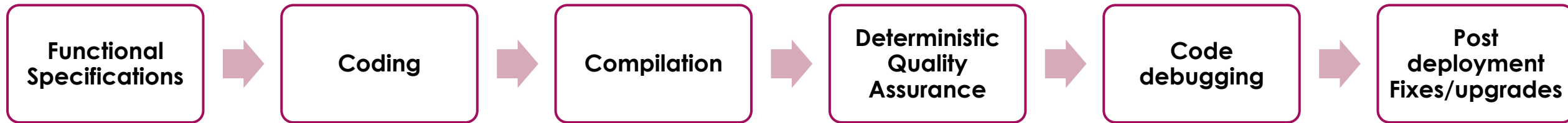
Traditional Programming Vs AI

How is AI Model different from Traditional Computer Programs ?

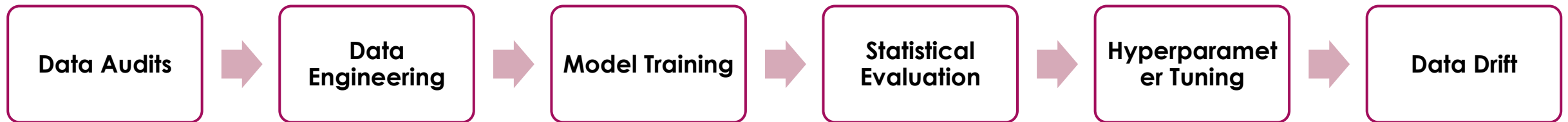
Traditional Computer Programs (Rule Based)	AI / Machine Learning Models (Learning Based)
Rule-based systems with fixed Logic. Works only for predefined conditions. Best for deterministic and repetitive tasks	Learning-based systems. Model behavior evolves through learning. Works for new unseen situations. Best for prediction, pattern recognition, and intelligent decision-making
Developers explicitly write logic and rules. Output depends on manually coded instructions	System learns patterns automatically from data. Output depends on learned relationships in data
Requires rule updates whenever scenarios change	Improves performance through training and more data
Difficult to handle highly complex real-world patterns	Excellent for handling large-scale complex datasets
Example: "If temperature > 40 → Raise alert"	Example: Predict machine failure using multiple sensor parameters
Traditional Flow: Data + Rules → Output	AI Flow: Data + Expected Outcomes → Trained Model → Prediction

Traditional Programming Vs AI

Traditional Software development (Writing rules, Code centric)



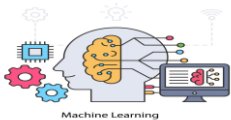
AI system Development (Learning from data, Data centric)



AI Vs ML Vs DL



Artificial Intelligence is broad term focused on creating systems that can perform tasks that typically require human intelligence.



Machine Learning is subset of AI that enables systems to learn from data and improve over time without being explicitly programmed.



Deep Learning is specialized subset of ML that uses **neural networks** with multiple layers to learn from large amounts of complex data.



Natural Language Processing understands, interpret, and respond to human language (text & speech) - chatbots, translation, sentiment analysis.

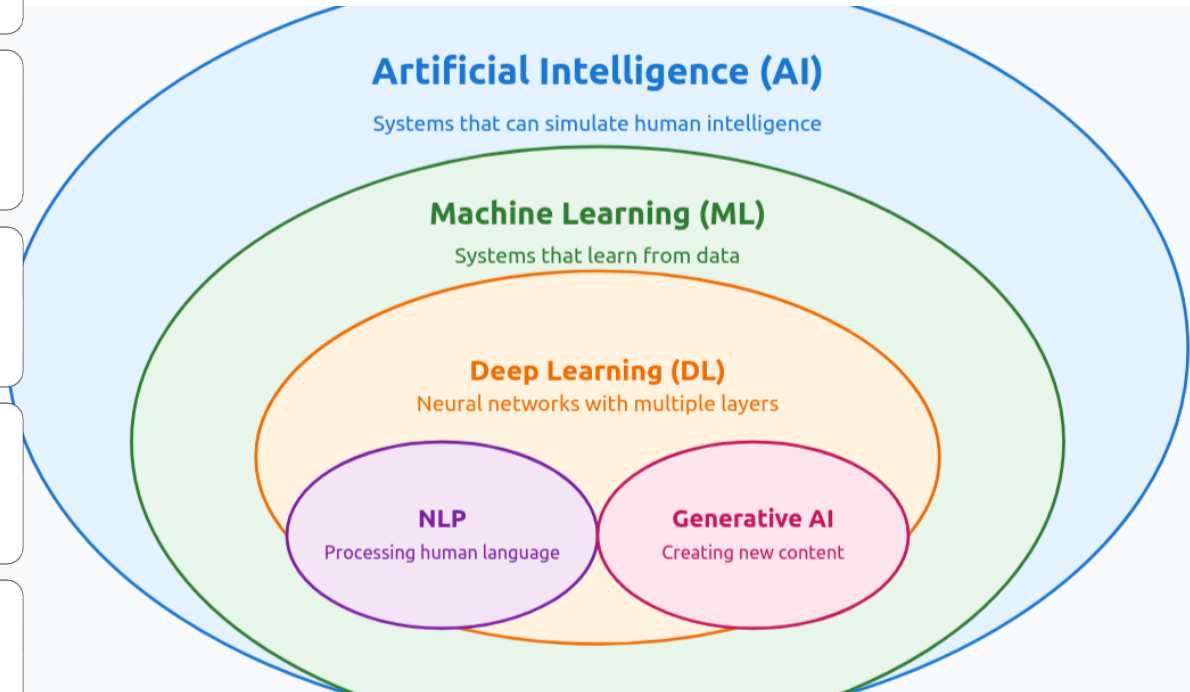


COMPUTER VISION

Computer vision -Helps computers understand and interpret visual data from the real world eg - object detection, facial recognition, image classification



Generative AI - AI that learns to generate data, content, or even entire solutions autonomously.



AI in IoT

Evolution of Connected and Intelligent systems



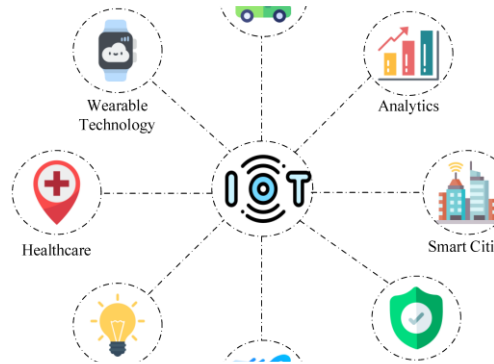
Internet

- systems became globally connected and information became digitally accessible



Mobile

- Computing moved into our hands through smartphones and mobile applications



IoT

- Not only humans but also devices, sensors, machines, and physical objects became connected to the internet



AIoT

- convergence of AI and IoT.
- Connected systems no longer just analyze, learn, predict, adapt, and make intelligent decisions in real time.